# Ulysse Hennebelle

Ph.D. Student in Operations Research
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## EDUCATION

**Cornell University - Cornell Tech** 2024-Present
*Ph.D. Student in Operations Research and Information Engineering* *New York, NY*
**Advisor:** Omar El Housni.
**Selected coursework:** Online Algorithms and Market Design, Mathematical Programming, Integer Programming, Statistical Principles, Scheduling Theory, Analysis of Algorithms, Applied Stochastic Processes, Convex Analysis, Randomized Algorithms.

**École Polytechnique** 2020-2024
*Ingénieur Polytechnicien (Bachelor's and Master's degree)* *Palaiseau, France*
**Final rank:** 34/435.
**Selected coursework:** Operations Research, Competitive Programming, Statistical Modeling of Processes, Deep Learning, Algebraic Topology, Quantum Physics.

## RESEARCH INTERESTS

Data-driven Optimization, Optimization under Uncertainty, Online and Stochastic Decision-Making, Combinatorial Optimization, Choice Modeling, Revenue Management, Assortment Optimization, Two-sided Marketplaces, Social Choice.

## PAPERS

### Published

[1] **Two-Sided Assortment Optimization: Adaptivity Gaps and Approximation Algorithms**
*with Omar El Housni and Alfredo Torrico.*
***Operations Research, 2026.***
Preliminary version: Adaptivity Gaps in Two-Sided Assortment Optimization, IPCO 2024.

[2] **Runtime Analyses of Multi-Objective Evolutionary Algorithms in the Presence of Noise**
*with Matthieu Dinot, Benjamin Doerr, and Sebastian Will.*
***Proceedings of IJCAI 2023, pp. 5549-5557.***

### Working Papers

[3] **Condorcet-Winning Sets and Peer Selection in Planar Metric Elections**
*with Gabriel de Azevedo.*
***arXiv:2608.27653, 2026.***

[4] **A Unified Framework for Assortment Optimization and Matching under MNL Choice**
*with Omar El Housni.*
***arXiv:2610.35389, 2026.***

## INDUSTRY & RESEARCH EXPERIENCE

**Data Scientist Intern** May-Aug 2026
*New York City Police Department | Siegel PiTech PhD Impact Fellow* *New York, NY*
- Built a Python pipeline for time series forecasting and spatial anomaly detection form four years of 6M NYC 311 and 17M NYPD 911 calls.
- Engineered rolling forecasts at 1-, 14-, and 28-day prediction horizons using 100+ history, calendar, location, and weather features. Benchmarked Machine Learning and time series models across three targets and three horizons. Optimized the best model to push R^2 above 0.53 on the hardest task.
- Assigned calls to census blocks within police precincts with GeoPandas and implemented Kulldorff's scan statistic to identify chronic locations with unusually high call volumes.
- Extended the scan statistic from circular regions to arbitrary convex shapes. Convex regions achieved a higher likelihood-ratio fit in 93% of comparisons. I optimized the implementation to obtain a 100x speedup.

**Research Assistant** Mar-Jul 2023
*Cornell Tech* *New York, NY*
- Developed optimization models and approximation algorithms for two-sided marketplaces with customer choice; work led to an Operations Research paper and an IPCO 2024 conference paper.

**Visiting Associate** Jun-Aug 2022
*Boston Consulting Group* *Casablanca, Morocco*
• Worked with BCG Digital Ventures on strategy and venture-building projects for large clients.

## TEACHING EXPERIENCE

Teaching assistant for graduate (M.Eng.) courses at Cornell Tech, serving classes of 60+ students.

**ORIE 5530 Modeling Under Uncertainty** Fall 2026
*Teaching Assistant, Graduate (M.Eng.)*

**CS 5854/ORIE 5138 Networks and Markets** Spring 2026
*Teaching Assistant, Graduate (M.Eng.)*

**ORIE 5132 Pricing Analytics and Revenue Management** Spring 2026
*Teaching Assistant, Graduate (M.Eng.)*

**ORIE 5530 Modeling Under Uncertainty** Fall 2025
*Teaching Assistant, Graduate (M.Eng.)*

**ORIE 5600 Financial Engineering with Stochastic Calculus** Fall 2024
*Teaching Assistant, Graduate (M.Eng.)*

## HONORS AND AWARDS

**Siegel PiTech PhD Impact Fellowship** Summer 2026
**Professor Robert E. Bechhofer Fellowship** Spring 2025

## TALKS

- INFORMS Annual Meeting, San Francisco, CA, 2026. **Online Matching with MNL Choices**.
- INFORMS Annual Meeting, Atlanta, GA, 2025. **Constrained Two-Sided Assortment Optimization**.
- Revenue Management and Pricing Conference, New York, NY, 2025. **Constrained Two-Sided Assortment Optimization.**

## LEADERSHIP AND SERVICE

**Voting Student Representative** 2021-2023
*Institut Polytechnique de Paris Board of Directors* *Palaiseau, France*
• Represented an 8,000-student body on the board of directors.

**Officer** Sep 2020-Apr 2021
*French Army* *France*
• Led a platoon of 28 recruits and managed military training, discipline, safety, and well-being.

## TECHNICAL SKILLS AND LANGUAGES

**Languages:** English (fluent); French (native).
**Programming and data:** Python, NumPy, pandas, scikit-learn, PyTorch, C++, SQL.
**Optimization:** Gurobi, CVXPY.
**Tools:** Git, LaTeX, Codex.
**Interests:** GeoGuessr (online geography game).